SC Honors Geometry - Nappier
4.3 – Rotations • Homework

Name: ____________________________________

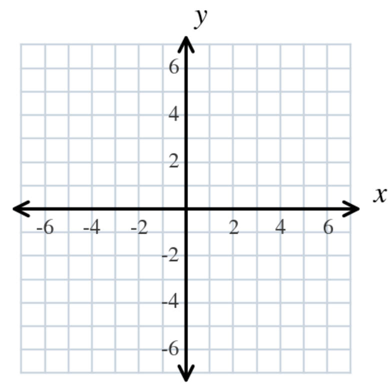
Find the image of A(4, −1) after each rotation about the origin.

1. 90°

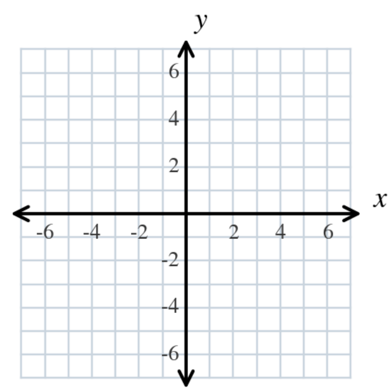
2. 180°

3. 270°

Use the coordinate planes at the right.



4. Graph △GHJ with G(2, 1), H(5, 3), and J(6, −1) and its image after a 90° rotation about the origin. List the image coordinates.



5. Graph △ABC with A(1, 1), B(3, 4), and C(5, 2) and its image after the composition below. List the final coordinates.
Reflection: in the x-axis   Rotation: 90° about the origin

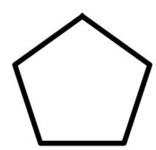
Find the coordinates of each image. You do not need to graph.

6. Quadrilateral KLMN with K(−4, 2), L(−1, 4), M(0, 1), and N(−3, −1) after a 180° rotation about the origin.

7. $\overline{PQ}$ with P(−2, 5) and Q(1, 3) after a 270° rotation about the origin.

8. $\overline{DE}$ with D(−5, 2) and E(−2, 4) after the composition:
Translation (x, y) → (x + 3, y − 6), then a 180° rotation about the origin.

Answer each question.



9. Does the regular pentagon at the right have rotational symmetry? If so, describe every rotation of 180° or less that maps it onto itself.

10. A student says that a 90° counterclockwise rotation and a 270° clockwise rotation about the origin give different images. Is the student correct? Use the point (3, 5) to explain.